RESEARCH · 03

The True *Contributors*.

WHO ACTUALLY MOVES THE TABLE? · EXPECTED POINTS PER 90 WITH HONEST RANKS

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renball

TL;DR

Three *sentences*

THE BOARD

Lionel Messi (0.81 EP per 90) and Kylian Mbappé (0.77) are the only players sure of a top-10 place. Number 3 could rank anywhere from 1 to 34.

WHAT REPEATS

EP per 90 carries over from season to season (r = 0.69); how decisive a player's goals are barely does (r = 0.17).

REALITY CHECK

The model names the actual award winner in 12 of 33 league-seasons (chance: 0.1); a goals + assists count names 17.

Price the goal, then *share* it

Goals win points, but not the same number of points, and a goal is rarely one player's work.

The price comes from Project 02: the expected final points just after a goal minus just before, from every minute of 13,206 matches. A go-ahead goal from level is worth 0.733 EP in the first 15 minutes and 1.507 in the last 15.

The credit goes to the scorer, SCA1 (the action before the shot) and SCA2 (the action before that), each at the goal's full value, once per player per goal: 8,694 role credits merge into 84,148 credits.

DATA

Five leagues, seven *seasons*

34,054	12,599	4,167	78.3%
GOALS PRICED	MATCHES	PLAYERS · 900+ MIN	GOALS WITH SCA2

FBref match pages, 2017–18 to 2023–24: Premier League, La Liga, Serie A, Bundesliga, Ligue 1. Shot tables, official scores, every player's minutes, position and age per match.

The data end in 2023–24: the SCA2 data end there because the source discontinued them.

Players told apart by birth date (45 shared names); minutes from the same matches as the goals.

METHOD

Shrink the noise, then *rank*

ADJUSTED RATE

Pulled toward the position average

EP per 90 is noisy over few minutes. Each rate is pulled toward its position's average, more when minutes are few: a forward with 1,000 minutes 79%, with 10,000 minutes 28%.

PLAUSIBLE RANK

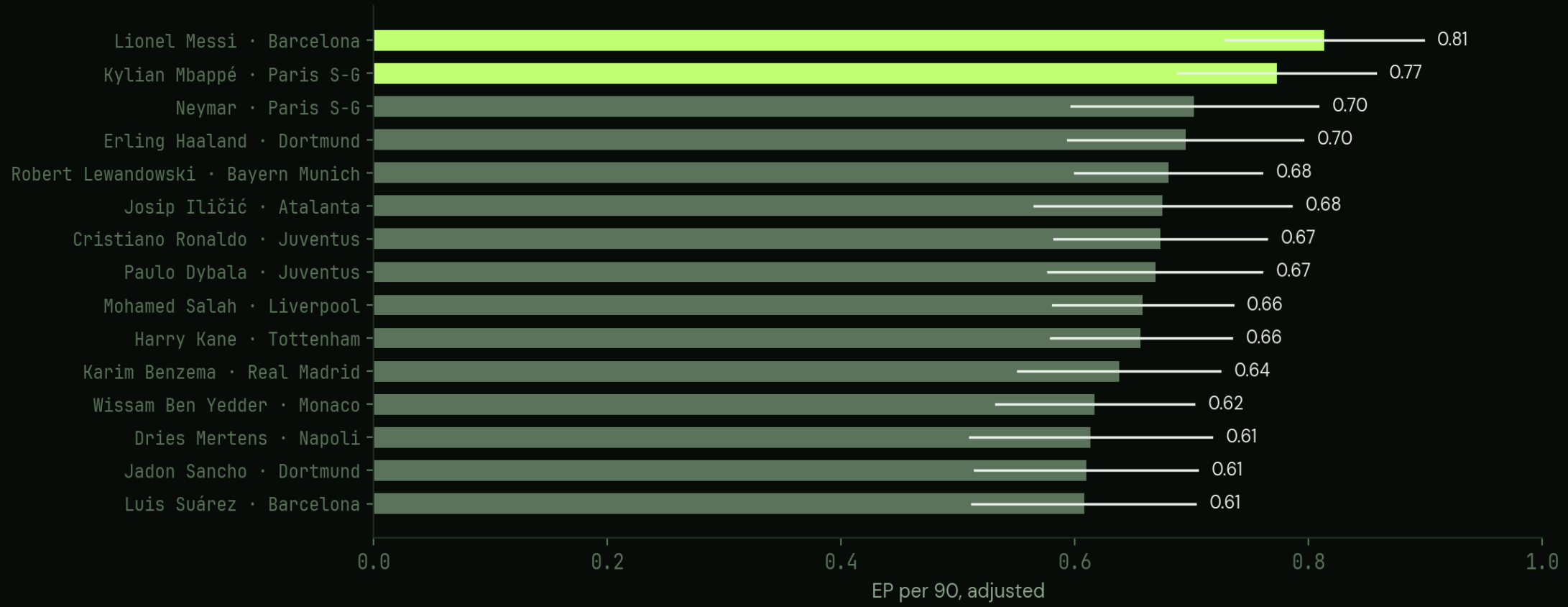
A range, not a place

Every player's adjusted rate is drawn 4,000 times with its uncertainty and ranked each time. The plausible rank is where he lands 95% of the time.

Empirical Bayes by position (DerSimonian-Laird priors) · measured ranges from Project 02's 1,000 match-bootstrap resamples · fixed seed

HEADLINE

Two players are *sure*. The rest is a shortlist.



4,167 players with 900+ minutes. Green: top-10 chance of 95% or more. Whisker: 95% interval of the adjusted rate.

THE BOARD

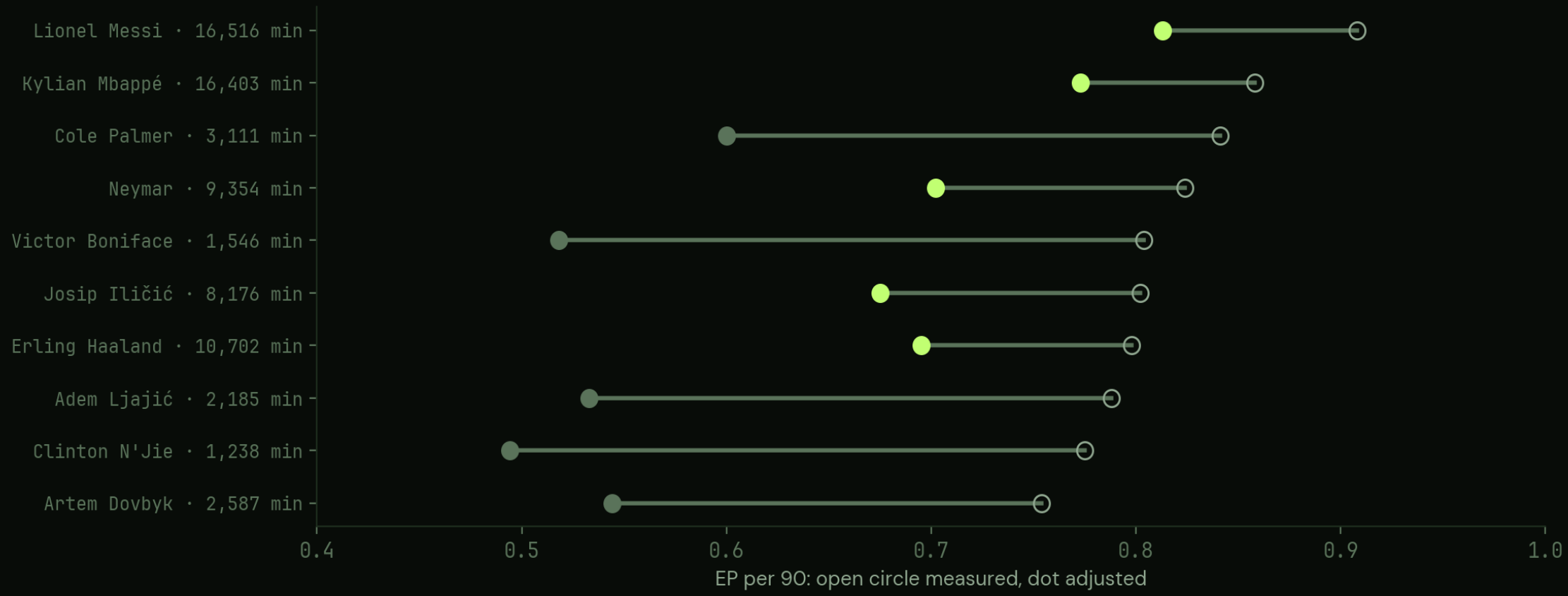
The top 10, with *plausible ranks*

#	PLAYER	TEAM	MINUTES	MEASURED	ADJUSTED	PLAUSIBLE RANK	TOP-10 CHANCE
1	Lionel Messi	Barcelona	16,516	0.908	0.813	1–4	100%
2	Kylian Mbappé	Paris S-G	16,403	0.858	0.773	1–7	100%
3	Neymar	Paris S-G	9,354	0.824	0.702	1–34	78%
4	Erling Haaland	Dortmund	10,702	0.798	0.695	2–31	75%
5	Robert Lewandowski	Bayern Munich	18,888	0.736	0.680	2–33	67%
6	Josip Iličić	Atalanta	8,176	0.802	0.675	2–57	60%
7	Cristiano Ronaldo	Juventus	13,704	0.748	0.673	2–41	59%
8	Paulo Dybala	Juventus	13,538	0.744	0.669	3–45	56%
9	Mohamed Salah	Liverpool	20,697	0.705	0.658	3–42	45%
10	Harry Kane	Tottenham	20,645	0.703	0.656	3–43	44%

EP per 90 measured and adjusted toward the position average. Plausible rank: 95% of 4,000 joint draws. Top-10 chance: share of draws in the top 10.

SHRINKAGE

The raw top 10 *misleads*



5 of the raw top 10 leave the adjusted top 10 (grey): all short careers. Cole Palmer's 0.841 over 3,111 minutes is pulled to 0.600.

ROBUSTNESS

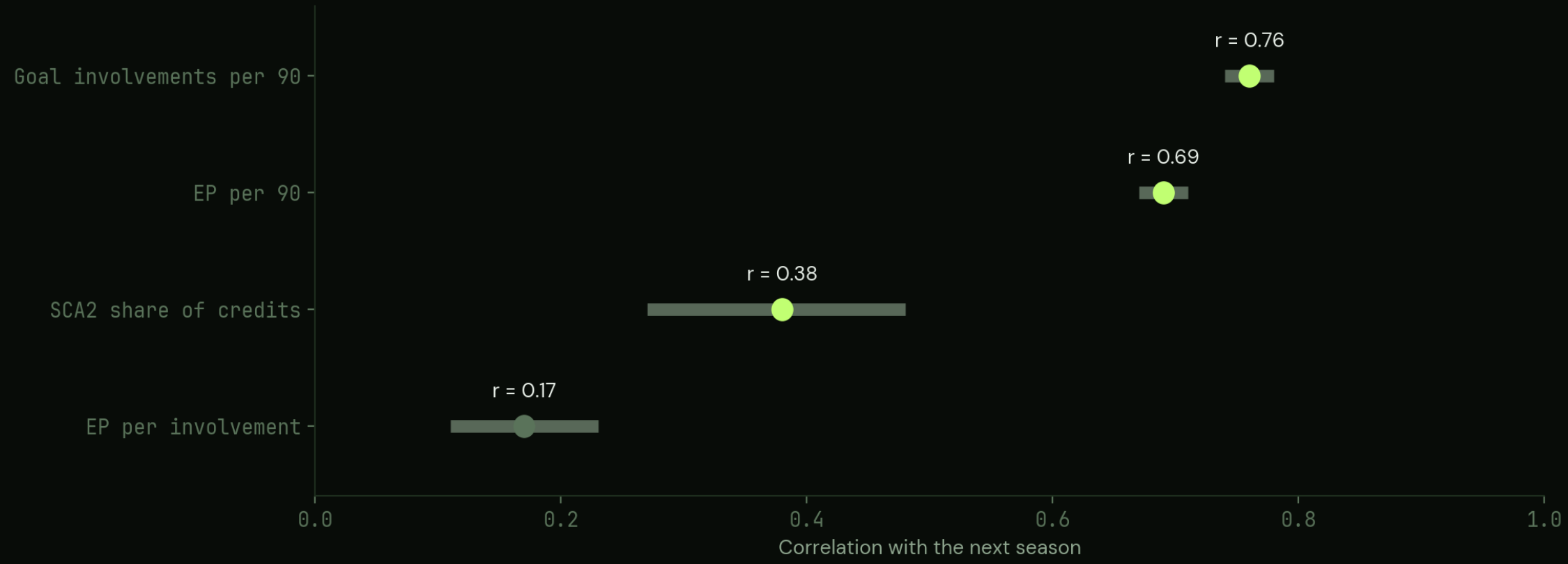
The credit rule barely *matters*

RULE VS FULL CREDIT	TOP-10 OVERLAP	RANK CORRELATION	LEADER
Every role a credit (old rule)	10	0.998	Lionel Messi
40/35/25 split	9	0.994	Lionel Messi

The split favours finishers: Mauro Icardi 39 → 17, Kevin De Bruyne 16 → 43.

SKILL OR NOISE

Volume repeats. Decisiveness *doesn't*.



Consecutive seasons, top 5 leagues. EP per 90: r = 0.69 (5,665 pairs); EP per involvement: 0.17. Lines: 95% intervals.

REALITY CHECK

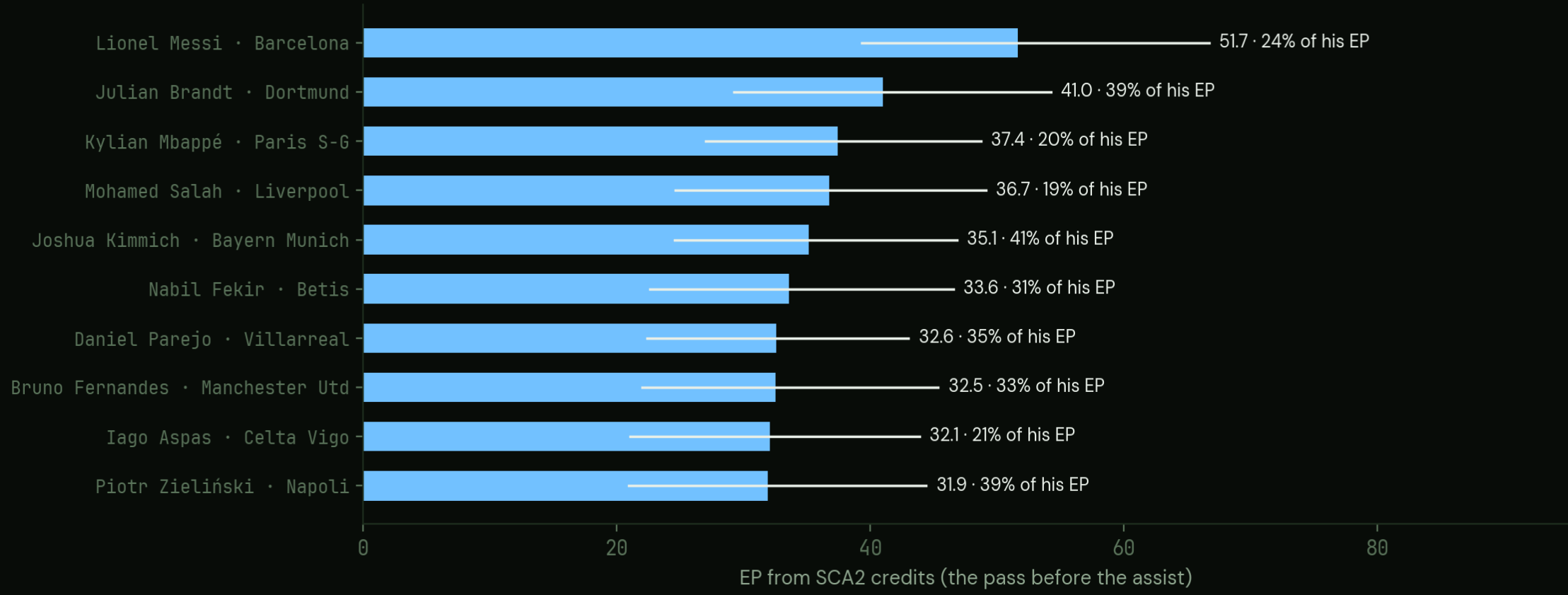
The model's pick vs the *award*

LEAGUE	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
Premier League	Salah · won	Agüero · #174	Mané · #4	Fernandes · #56	Salah · won	Haaland · won	Palmer · #10
La Liga	Messi · won	Messi · won	Messi · #2	Messi · #6	Benzema · won	Griezmann · GK winner	Sørloth · #3
Serie A	Dybala · #7	Mertens · #4	Ronaldo · #3	Muriel · #2	Immobile · #5	Osimhen · #2	Dybala · #8
Bundesliga	Lewandowski · won	Sancho · #4	Sancho · #2	Lewandowski · won	Schick · #4	Nkunku · #70	Guirassy · #7
Ligue 1	Fekir · #2	Mbappé · won	Neymar · no award	Mbappé · won	Mbappé · won	Messi · #3	Mbappé · won

Pick (surname) · won: the pick won the award · #: the actual winner's rank on the model's list. 12 of 33 named; top 3 in 20; chance 0.1; goals + assists leader 17.

SCA2

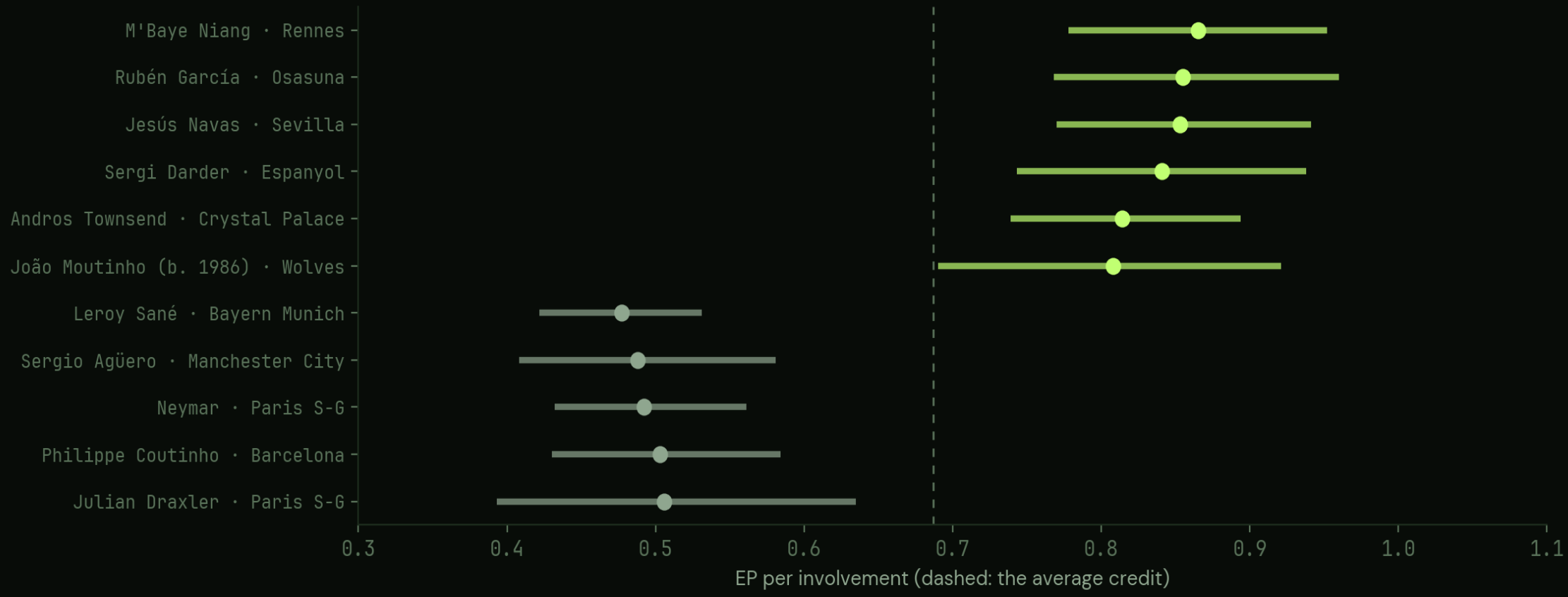
The hidden *architects*



Julian Brandt is second with 40.98 EP from SCA2 alone, but his plausible rank runs from 1 to 26. Roles counted separately; whiskers: 95%.

EP PER INVOLVEMENT

Decisive goals describe the *team*



Average credit 0.687 EP. 26 players clearly above, 53 below (chance: ~11 each). The lowest all played for Barcelona, Bayern Munich, Manchester City or Paris S-G.

POSITIONS

Best by *position*

#	FORWARD	ADJ.	MIDFIELDER	ADJ.	DEFENDER	ADJ.
1	Lionel Messi	0.813	Kevin De Bruyne	0.604	Raphaël Guerreiro	0.389
2	Kylian Mbappé	0.773	Andrej Kramarić	0.591	Achraf Hakimi	0.351
3	Neymar	0.702	Thomas Müller	0.571	Yannick Carrasco	0.348
4	Erling Haaland	0.695	Marco Reus	0.570	Filip Kostić	0.345
5	Robert Lewandowski	0.680	Bruno Fernandes	0.539	Álex Grimaldo	0.329

Each pulled toward its own position's average (forwards 0.402, midfielders 0.221, defenders 0.114). Goalkeepers left out: best 0.012.

23 OR UNDER

Young players to *watch*

#	PLAYER	TEAM	AGE	MINUTES	MEASURED	ADJUSTED	OVERALL RANK
1	Erling Haaland	Dortmund	23	10,702	0.798	0.695	4
2	Cole Palmer	Chelsea	22	3,111	0.841	0.600	19
3	Jamal Musiala	Bayern Munich	21	6,302	0.621	0.533	51
4	Bukayo Saka	Arsenal	22	13,392	0.556	0.522	60
5	Vinicius Júnior	Real Madrid	23	11,579	0.560	0.521	62
6	Victor Boniface	Leverkusen	23	1,546	0.804	0.518	69
7	Khvicha Kvaratskhelia	Napoli	23	5,255	0.587	0.509	79
8	Lucas Beltrán	Fiorentina	23	1,692	0.712	0.497	93

Aged 23 or under on 30 June 2024 (end of the data), 900+ minutes: 505 players. Short careers are pulled hardest.

TAKEAWAYS

Who should *care*

Coaches

TRAINING & TACTICS

No defensible claim for this audience from this study.

Managers

MATCH-DAY DECISIONS

No defensible claim for this audience from this study.

Clubs

RECRUITMENT & CONTRACTS

STRONG

Rank adjusted rates, not raw per-90s: 5 of the raw top 10 are short careers that drop out.

4,167 players · empirical Bayes by position

STRONG

EP per 90 repeats (r = 0.69); decisive-looking goals barely do (r = 0.17). Pay for involvement.

95% CI 0.67–0.71 and 0.11–0.23

Agents

POSITIONING A PLAYER

MODERATE

A pre-assist habit is partly real: the SCA2 share repeats (r = 0.38). A fair line in a playmaker's case.

352 season pairs · 95% CI 0.27–0.48 · SCA2 on 78% of goals

EXPLORATORY

Awards follow box scores: the goals + assists leader matched 17 of 33 winners, the EP model 12.

five different awards · a count, not a test

EVIDENCE STRONG MODERATE EXPLORATORY Definitions: methodology § Evidence labels

LIMITATIONS

What this *can't* tell you

- Prices, not causes: a goal's value says how much it mattered, not how hard it was.
- Three roles only: defending, pressing and off-ball runs earn nothing; keepers and defenders are undervalued.
- No team or league adjustment: players at high-scoring clubs get more involvements.
- SCA2 missing for 21.7% of goals; the data end in 2023-24 (source discontinued).
- A normal prior pulls true outliers in too hard; plausible ranks ignore shared price uncertainty.

CONCLUSION

What this *changes*

Two certainties. *Then a shortlist.*

Lionel Messi and Kylian Mbappé are clear of everyone at 0.81 and 0.77 EP per 90. Below them, ranks are ranges. Short careers inflate raw per-90s, the volume of a player's involvement repeats while the decisiveness of his goals does not, and award voters follow the box score. For a club, the useful number is the adjusted rate with its plausible rank.

Thanks for reading.

Full methodology, source code and data:

renball.com

```
/PROJECTS/03-TRUE-CONTRIBUTORS/

→ methodology      the shrinkage, the rank intervals, the award check

→ dashboard        every player and player-season, with intervals

→ code/            players.py · shrink.py · build.py
```